midflight and demanded a fork. The waiter nodded, walked over to another table piled high with dirty plates and assorted garbage, and looked over the scraps for a possible fork. We yelled out, "Never mind," and the diner who was short a fork made do with his fingers.

19

With my seam-sealer project completed in the Navy Yard, I went on to handle a dye-marker project.

A dye marker is a reddish-brown powder containing fluorescein which, when dumped in water, produces a bright and fluorescent green solution. It is kept in a plastic container that can be ripped open. If an airman is forced to leave his plane over sea and to float about on a gas-filled rubber raft, he might be able to stay afloat and alive for days, but he would be difficult to spot from potential rescue planes high in the

8 See The Early Asimov.

9 See Foundation and Empire (Gnome Press, 1951, later published by Doubleday). It appears in the book as "Part I—The General."

sky. By dumping the dye marker in the ocean when he sees a friendly plane overhead, a splotch of bright green is produced that is far easier to see than the boat itself.

The great task I set myself was to work out some way of devising a laboratory test that would predict, accurately, how well the dye marker would be seen from a plane. The sure way of doing it would be to have someone fly in a plane and note the ease with which different samples of dye marker could be noted. That, however, it seemed to me, would be very expensive in terms of planes, gasoline, and pilots—to say nothing of the chance of accident—so that a satisfactory laboratory test would be valuable indeed.

The only trouble was that in order to decide whether the laboratory test gave useful results, they would have to be checked from the air to begin with. Working with a batch of different samples of dye markers, you would run the lab test on each and list the dye markers in order of efficiency. You would then view them all from the air and check whether they were in the same order of efficiency. You could then check the boundary limit of effectiveness for sighting and you would be all set. No one would then ever have to fly a plane to check a dye marker again.